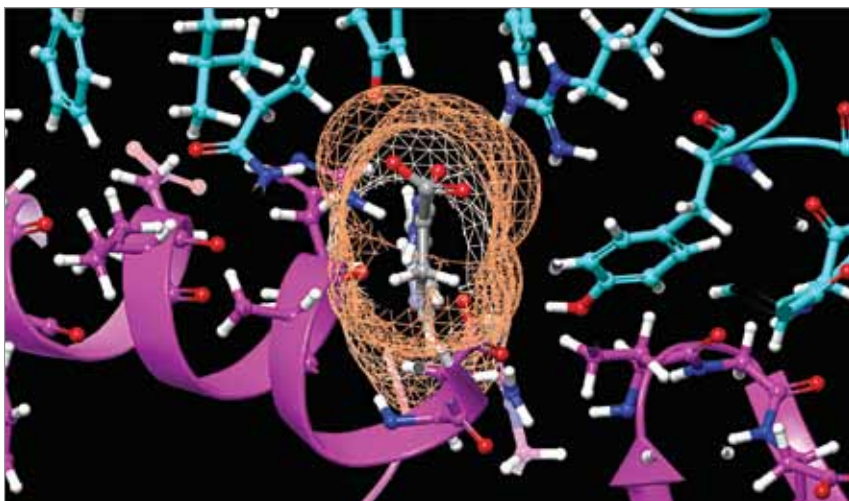




A compound, shown in gray, is calculated using supercomputers to bind to the spike protein and prevent it from docking to the Human ACE2 (Credit: <https://spectrum.ieee.org>)

Structure of Covid-19

The source of this unknown respiratory infection is related to a novel coronavirus that caused outbreaks of SARS from 2002 to 2004. The virus is called SARS-CoV-2, and the World Health Organization has named it Covid-19 (coronavirus detected in Dec 2019). Covid-19 particles are spherical in shape and have protein spikes protruding from the surface. These spikes latch onto human cells and undergo a structural change such that the viral membrane gets fused with the human cell membrane. The viral genes then enter the host cell producing more viruses. Covid-19 has four structural proteins, envelope (E), spike (S), membrane (M), and nucleocapsid (N).

The S, M, and E proteins form the envelope of the virus. The S-proteins give the virus a crown-like appearance, thus the name coronavirus. The M protein, the most abundant, is responsible for the shape of the envelope. The E protein, the smallest, is the structural protein. Both the S and M proteins help in virus assembly during replication. N protein is largely involved in the replication cycle (assembly and budding) and the host cellular response to viral infection. The Covid-19 spikes, which are ten to twenty times more likely to bind human cells compared to the 2002 SARS virus spikes, enable it to spread more easily from person to person.

In spite of similarities in the sequence and structure of the spikes of the two viruses, the 2002 SARS virus antibodies do not successfully bind to the Covid-19 spike protein, indicating that the vaccine and antibody-based treatments have to be unique to this new virus. Researchers are now experimenting with the spike protein to isolate antibodies from people who have recovered from infection to treat new infections before a vaccine becomes available. Cultured cells are used by scientists to make large quantities of the protein available for analysis. Cryo-electron microscopy is used to take pictures of the spike protein, and the images are combined to construct a 3D view of the virus.

Application of IT in fighting Covid-19

Supercomputers. The scientific community needs to study how the proteins' spikes attach to the wall of a human cell, how does the RNA of the virus get inside the human cell, and how does it take over the human DNA machinery. Large numbers of these experiments, when conducted in the lab, take a long time to be completed.

Simulations of experiments using supercomputers help biologists and scientists to find treatments and vaccines in the shortest time possible. The present-day supercomputers function at a speed as high as 'peta'—a level higher than 'tera' (the levels in ascending order are kilo, mega, giga, tera, peta, and exa). These are

a powerful resource for simulation of biochemical processes, materials, and chemical agents and help process a massive number of calculations as well as solve a vast number of complex and complicated equations related to bioinformatics, epidemiology, and molecular modelling. As the Covid-19 cases are increasing at a very high rate, a huge amount of data is available to researchers to find a drug that could work against the novel coronavirus.

Supercomputers look through the databases of existing drug compounds to identify drug compounds and molecules that have the highest binding affinity with the virus to disarm it most efficiently. These supercomputers also study the structure of the virus and how it interacts with cells in the human body as well as analyse the spread of the virus in a population.

In a global effort to tackle the pandemic, a number of countries are employing supercomputers to expedite their research programmes. A joint public-private effort, which includes IBM, Google, Amazon, MIT, Carnegie Mellon, and NASA called 'Covid-19 High-Performance Computing Consortium,' has been launched in the US. Japan is using the Fugaku supercomputer (successor to the 'K' computer, which had the highest calculation speed in the world).

China is reported to be using the Tianhe-1 supercomputer to diagnose Covid-19 patients from chest scans.

In India, the Centre for Development of Advanced Computing (C-DAC) is collaborating with the National Institute of Virology (NIV), IITs, Indian Institutes of Science Education and Research (IISER), labs under the Council of Scientific and Industrial Research (CSIR), Department of Biotechnology (DBT), as well as industries and startups for drug repurposing simulations towards discovering a suitable and effective drug for Covid-19. C-DAC, a part of the Indian Institute of Science (IISc), has launched a hackathon SAMHAR-Covid-19 to fight the coronavirus. (SAMHAR stands for 'Supercomputing using AI, ML, Healthcare Analytics based Research, and it means 'to vanquish' in Sanskrit.).

The hackathon is open to research-